

NVIDIA Nemotron 3 Nano Omni: One Open Model for Vision, Audio, and Text at 9x the Throughput

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The graphic features a dark blue background with a subtle grid and a faint line graph in the top right corner. The main title is in large, bold, white and teal text. Below it, a subtitle in smaller white text describes the model as an 'Open 30B-A3B hybrid MoE unifying multimodal agent perception'. Three key metrics are highlighted in separate boxes with vertical bars: '9.2x' in orange for throughput, '3B' in yellow for active parameters, and '6' in blue for leaderboards. The date 'May 11, 2026' is at the bottom left, and the 'ToKnow.ai' logo is at the bottom right.

Nemotron 3 Nano Omni: Vision, Audio, and Text at 9x Throughput

Open 30B-A3B hybrid MoE unifying multimodal agent perception

- 9.2x**
Higher throughput vs open omni models (video)
- 3B**
Active params per token (of 30B total MoE)
- 6**
Leaderboards topped for docs, video, and audio

May 11, 2026

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NVIDIA released [Nemotron 3 Nano Omni](#) on April 28, a 30-billion-parameter open multi-modal model that handles text, images, audio, and video in one architecture. The “30B-A3B” label means 30 billion total parameters but only about 3 billion active per token, thanks to

a hybrid Mixture-of-Experts design that blends [Mamba layers](#) (for memory-efficient sequence processing) with transformer layers (for precise reasoning). NVIDIA reports [9.2x higher system throughput](#) for video tasks and 7.4x for multi-document tasks compared to other open omni models at the same per-user responsiveness threshold. It tops six leaderboards for document intelligence, video understanding, and audio comprehension, including [OCRBenchV2](#) and [VoiceBench](#). The model ships with open weights, training data, and full recipes, and runs on everything from [Jetson edge hardware](#) to cloud GPUs via [Ollama](#), vLLM, and 25+ partner platforms.

The payoff is collapsing three or four separate models (vision, speech, language, document parsing) into one. An AI agent that needs to watch a screen recording, read a chart, and listen to a voice note can now do it in a single inference loop instead of chaining separate models. [H Company’s computer-use agent](#), built on Nano Omni, processes full 1920x1080 screen recordings in real time. Since only 3B parameters fire per token, the model fits on hardware that a dense 30B model never could.

Multimodal AI is moving from “bolt separate encoders onto a language model” to unified architectures where vision, audio, and text share the same reasoning loop. [LongCat-Next](#) showed a similar approach at 74B total (3B active), and now NVIDIA’s entry brings full-stack deployment support from edge to cloud.

Sources:

- [Nemotron 3 Nano Omni Technical Report \(arXiv\)](#)
- [NVIDIA Technical Blog: Nemotron 3 Nano Omni](#)
- [NVIDIA Blog: Launch Announcement](#)
- [Nemotron 3 Nano Omni on Hugging Face](#)

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